

Design Basis Accident: Military Action Against Nuclear Facilities

DBA-MA L2 — IAEA Accident Description

Formal Accident Description and Design Requirements

Version 3.0

September 2026

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T. Roxey, “Design Basis Accident: Military Action Against Nuclear Facilities — Formal Accident Description and Design Requirements,” Version 3.0, Eclectic Technologies, September 2026.

DOI: <https://doi.org/10.5281/zenodo.22945300>

Current version: https://scubanuke.github.io/publications/dba-ma/L2_DBA-MA_IAEA_Accident_Description.pdf

A Child Document Under the DBA-MA International Framework

1. Scope and Applicability

1.1 Purpose

This document establishes the Design Basis Accident language for military action against nuclear power facilities (DBA-MA). It defines the postulated accident, the conditions it creates, the required safety endpoint, and the engineering and operational provisions required for mitigation. This DBA supplements, and does not replace, any existing Design Basis Accident in the facility's licensing basis.

This document is an international reference document within the DBA-MA series. It applies to all national implementations that adopt the DBA-MA framework. National implementations will map the requirements of this document to their specific regulatory structure, facility inventory, and threat environment, as set out in the corresponding national child documents.

1.2 Applicability

The DBA-MA applies to all commercial nuclear power facilities operating under applicable national regulatory requirements within the scope of the DBA-MA framework. The specific engineering and operational requirements will vary by reactor type, site configuration, and licensing basis. However, the accident description, safety endpoint, and general design requirements established in this document apply to all facilities within scope. Other reactor types and facility classes are addressed through facility-class instruments, beginning with small modular reactors in DBA-MA-SMR-FC1; the principles of this document apply to other nuclear installations by analogy, as the parent framework sets out in its Section 1.

National child documents define the specific regulatory instruments, authority structures, and facility inventories to which this international document applies within each jurisdiction. National-specific acceptance criteria and regulatory references are addressed in the applicable national child document, not in this document.

1.3 Relationship to Existing Design Basis

The DBA-MA represents an envelope case that encompasses multiple existing DBAs occurring concurrently, with the additional conditions of intentional causation, sustained duration, adaptive threat behavior, and simultaneous compromise of external support systems. Engineering features designed to satisfy the DBA-MA will provide additional margin for all existing individual DBA scenarios.

This document uses the language of the design basis accident deliberately. It adopts the discipline of design basis analysis: defined conditions to be withstood, credited structures, systems and components, and explicit acceptance criteria, with the ASSC as the safety endpoint. Its proposed placement in the IAEA standards hierarchy is as a category of design extension

condition, as the parent framework explains in its Section 8.1. The analytical method is that of the design basis; the regulatory home is the design extension condition.

1.4 Classification

The general accident description, design principles, safety endpoint definition, and non-specific engineering requirements contained in this document are suitable for public release. Specific threat characterizations, vulnerability assessments, hardening specifications, and details of site-specific mitigation measures may require handling under national security classification in accordance with applicable national law. They will be documented separately through appropriate channels. This parallels the existing approach to the Design Basis Threat in international nuclear security practice.

Classification handling for national implementation supplements is governed by the applicable national regulatory and security framework, as addressed in each national child document.

2. Accident Description

2.1 General Description

The Design Basis Accident for Military Action (DBA-MA) postulates that a nuclear power facility is subjected to military operations by a hostile nation-state or state-equivalent force. The military action may include seizure and occupation of the facility, direct application of explosive ordnance against plant structures and systems, deliberate disruption of off-site electrical transmission infrastructure, interdiction of supply chains and transportation routes serving the facility, interference with or displacement of operating staff, and disruption of communications with regulatory authorities, emergency response organizations, and corporate support. These conditions correspond to the six threat categories established in the parent framework, Section 2.5: seizure and occupation, bombardment by explosive ordnance, uncrewed aerial systems, cyber attack, denial of external support, and electromagnetic attack. An actual attack will usually combine several of them.

The DBA-MA differs from all existing DBAs in three fundamental respects. First, the initiating conditions are imposed by an adaptive adversary that can adjust tactics in response to defensive and mitigating actions, rather than by natural phenomena, equipment failures, or human error. Second, the accident conditions are sustained indefinitely rather than progressing through a discrete event-response-recovery sequence. Third, the external support systems upon which existing safety analyses depend for recovery—off-site power, emergency response, supply chains, regulatory oversight, staffing pipelines—are simultaneously compromised by the same event that challenges the facility's safety functions.

2.2 Phased Accident Progression

Unlike existing DBAs, which assume a discrete initiating event, the DBA-MA is structured as a phased accident progression that reflects the operational reality of military action. The facility's response obligations and the applicable engineering and operational requirements differ across phases.

2.2.1 Phase 1: Preparation (Left of Boom)

Phase 1 encompasses the period during which indicators of potential military action against the facility or its region are observed, but no attack has occurred. This phase may extend from weeks to months. During Phase 1, the facility remains in normal operation and continues to generate electricity in support of the national grid and defense needs. Concurrently, the facility conducts preparation activities to position the plant for rapid transition to the Autonomous Safe Shutdown Condition, if required.

The operational dilemma inherent in Phase 1 is explicitly recognized and addressed in this DBA: the facility is simultaneously obligated to maintain electrical generation to support national defense and civilian needs, and to prepare for the possibility that continued operation will become untenable. The resolution of this dilemma is governed by the Trigger Framework (DBA-MA L3, companion document), which establishes observable criteria for transitioning from continued operation to a controlled shutdown.

Phase 1 engineering and operational requirements include maintenance of consumable inventories (diesel fuel, lubricants, water treatment chemicals, and other materials necessary for extended autonomous operation) at levels specified for the DBA-MA; verification of readiness of all systems designated for autonomous safe stable state operations; pre-positioning of emergency equipment and materials that will be required after loss of external supply chains; exercise of military action response procedures with operating staff; establishment and verification of communication channels with intelligence, military command, and emergency management authorities as specified in the Trigger Framework; and preparation of the facility for accelerated cooldown and transition to safe shutdown from any operating state.

2.2.2 Phase 2: Transition

Phase 2 begins when the trigger criteria specified in the companion Trigger Framework (DBA-MA L3) are met, triggering the decision to initiate a controlled shutdown and transition to the Autonomous Safe Shutdown Condition. Phase 2 ends when the facility has achieved the Autonomous Safe Shutdown Condition as defined in Section 3 of this document.

The transition must be accomplishable under deteriorating conditions that may include degradation of off-site power (intermittent or lost), reduced staffing levels, partial loss of communications, and the early effects of military action in the facility's region. The transition procedure must be executable from any initial operating state (full power, reduced power, hot standby, or hot shutdown). It must be designed to be completed within a timeframe that provides a margin against anticipated conditions that could prevent completion.

The transition is analogous to the controlled shutdown executed by facilities in hazard-prone areas when sustained conditions are predicted to exceed design-basis thresholds. It is extended to account for the specific conditions of the military action scenario. The trigger criteria, authority for the shutdown decision, and required completion timeframes are specified in the companion Trigger Framework document.

2.2.3 Phase 3: Autonomous Safe Shutdown Condition

Phase 3 begins when the facility achieves the Autonomous Safe Shutdown Condition as defined in Section 3 and continues for the duration of the military action. This phase may extend for months to years. During Phase 3, all safety functions are maintained using only systems, equipment, materials, and personnel physically present at the facility site. No dependence on external systems—off-site power, supply chains, communications, emergency response, regulatory oversight, or staff rotation—is assumed or required.

Phase 3 represents the sustained endurance condition for which the facility's DBA-MA engineering features are designed. The design must ensure that the Autonomous Safe Shutdown Condition can be maintained indefinitely, with time margins that increase rather than decrease as decay heat declines.

2.2.4 Phase 4: Recovery

Phase 4 begins when the military situation changes enough that external support can be restored to the facility. This may result from a military withdrawal, liberation, a ceasefire, or another resolution of hostilities. Recovery from the DBA-MA is categorically different from recovery following any existing DBA, because the facility may have been under hostile occupation for an extended period, during which systems have degraded without maintenance, documentation and records may be incomplete or compromised, physical modifications or damage by the occupying force may be undocumented, and the facility condition may be unknown to the recovering operator.

Phase 4 requires a comprehensive reassessment of the facility before any restart activities are considered. The scope of this reassessment is comparable to a post-accident recovery evaluation, not a return-to-service inspection following a normal transient. The fundamental requirement is that no restart activities shall commence until a comprehensive assessment demonstrates that all safety systems meet their original design and licensing requirements or that appropriate compensatory measures have been established.

2.2.4.1 Phase 4 Triggering Conditions

Phase 4 is initiated when at least one of the following conditions is established: (a) military forces hostile to the facility have withdrawn from the site and its immediate access perimeter; (b) national or allied military authority has certified that the facility is accessible for recovery operations; or (c) a designated third-party inspection authority — IAEA or equivalent — has been granted and exercised unimpeded access to the site. No single condition is sufficient if the

others indicate a continuing threat. The authority responsible for making the Phase 4 initiation determination is specified in the applicable national child document.

2.2.4.2 Comprehensive Facility Reassessment

Before any restart or return-to-service activities commence, the facility shall complete a comprehensive reassessment addressing the following elements in sequence.

Physical condition survey. A systematic survey of all safety-related structures, systems, and components shall be conducted to identify damage, unauthorized modification, equipment repositioning, and any condition inconsistent with the pre-occupation configuration. Qualified nuclear safety engineers shall survey with access to the pre-occupation design basis documentation. Where pre-occupation records are unavailable or of uncertain integrity, the as-found condition shall be treated as the baseline and compared against original licensing drawings.

Undisclosed modification protocol. The occupying force may have installed, removed, repositioned, or modified systems and equipment without documentation. The reassessment shall include a systematic search for undisclosed modifications using physical inspection, radiographic techniques where appropriate, and comparison against the pre-occupation configuration at the component level for all safety-related systems. No safety-related system shall be returned to service until its configuration has been verified against the original design basis or a deviation documented and dispositioned under the applicable safety review process.

Radiological characterization. The radiological condition of the facility shall be characterized before personnel access is extended beyond radiation protection controls. Areas where containment may have been breached, where fuel damage may have occurred, or where occupying forces may have disturbed radioactive materials shall receive priority characterization.

Safety culture and personnel assessment. Personnel who remained at the facility during the occupation shall receive a structured debriefing covering operational decisions made during Phase 3, any coerced actions, the current status of safety systems as understood by operating staff, and any anomalies or concerns the staff were unable to document or report during the occupation. The reassessment shall include an evaluation of the fitness of the returning operating organization to resume licensed activities.

Documentation reconstruction. Operating records, maintenance records, modification records, and the facility's licensing basis documentation shall be reviewed for completeness and integrity. Where records are missing, altered, or of uncertain provenance, the affected systems shall be treated as uninspected and subjected to the physical condition survey protocol regardless of apparent condition.

2.2.4.3 Regulatory Authority Structure for Phase 4 Oversight

The national regulatory authority bears primary responsibility for authorizing Phase 4 activities and the eventual return to service. Where the national regulatory authority's own infrastructure has been degraded by military action, the applicable national child document shall specify the interim authority arrangement, which may include IAEA technical oversight in an advisory role pending restoration of national regulatory capacity. No restart activities shall be authorized under an arrangement that lacks independent technical review capability, regardless of commercial or grid-reliability pressure to restore generation.

2.2.4.4 Sequencing of International and National Authorization

IAEA inspection and assessment under Phase 4 shall precede national regulatory authorization for restart where the facility was under occupation by a foreign state. The IAEA assessment shall address: confirmation that the facility's nuclear material inventory is consistent with pre-occupation records; identification of any conditions that constitute an immediate radiological hazard requiring remediation before national regulatory review proceeds; and a preliminary finding on the adequacy of the physical condition survey. The IAEA assessment is advisory to the national regulatory authority and does not substitute for the national licensing determination.

2.2.4.5 Phase 4 Procedures

Each facility shall develop and maintain Phase 4 recovery procedures as required under Section 5.1. These procedures shall be integrated with the IAEA safeguards re-establishment process and the applicable national regulatory return-to-service requirements, and shall address the specific conditions — degraded documentation, potential undisclosed modifications, personnel continuity gaps — that distinguish Phase 4 recovery from a normal post-outage return to service.

3. Autonomous Safe Shutdown Condition

3.1 Definition

The Autonomous Safe Shutdown Condition (ASSC) is a plant condition in which the reactor is shut down, decay heat is being removed through systems that do not depend on external inputs, spent fuel is in a safely cooled and monitored condition, containment integrity is maintained, and the facility can remain in this condition indefinitely using only resources physically present at the site. The ASSC is the required safety endpoint for the DBA-MA.

The ASSC is distinct from the cold shutdown defined in the current Technical Specifications under national regulatory frameworks. A cold shutdown is a transient condition on the path to either a restart or maintenance, with the implicit assumption that external support systems remain available. The ASSC is an endpoint condition designed to last indefinitely without external support. It is the condition the facility is designed to achieve and maintain until the military

situation permits the restoration of external support and the commencement of recovery activities.

3.2 Required Characteristics

A facility in the Autonomous Safe Shutdown Condition must satisfy the following conditions simultaneously and must be capable of maintaining all of them for an indefinite duration.

Subcriticality. The reactor must be maintained subcritical with an adequate shutdown margin. Subcriticality must be assured by passive means (control-rod insertion, soluble boron, and burnable absorbers) that do not require active system operation or operator action to be maintained.

Decay heat removal. Decay heat must be removed at a rate that maintains the integrity of the fuel cladding and the reactor vessel within the limits of the analysis. The decay heat removal pathway must be achievable through passive means or active systems capable of operating on stored energy (e.g., diesel fuel, batteries) for the duration of the ASSC, with design provisions that ensure time margins increase as decay heat declines. The design basis must account for entry into the ASSC at the earliest credible time following reactor shutdown, corresponding to the maximum decay heat load.

Spent fuel pool cooling. Spent fuel stored in the spent fuel pool must be maintained in a safely cooled condition. Cooling must be achievable without dependence on makeup water from external sources for the duration of the ASSC. Design provisions must account for the maximum fuel-loading condition, including, if applicable, a core that has recently been offloaded.

Containment integrity. The containment structure must maintain its pressure boundary function. Containment isolation must be achievable and maintainable without active system support beyond that available from on-site resources. The design must account for the possibility that containment penetrations or structures may sustain damage from explosive ordnance.

Radiological monitoring. The facility must be capable of monitoring radiological conditions within the containment, in the spent fuel pool area, and at the site boundary using instrumentation powered by stored energy that does not require calibration or maintenance for the duration of the ASSC.

Essential instrumentation. Minimum instrumentation necessary to confirm that the ASSC conditions are being maintained—reactor coolant temperature and pressure, spent fuel pool level and temperature, containment pressure and radiation levels, and stored energy status (diesel fuel level, battery charge)—must be available and functional throughout the ASSC duration.

3.3 Design Basis Duration

The ASSC must be maintainable for a minimum design-basis duration. That minimum is set by facility class, as described below; site-specific analysis and national regulatory review may

extend it but may not reduce it. The design-basis duration must account for the demonstrated duration of military occupations of nuclear facilities, which the Ukrainian precedent has shown can extend for years. Systems that depend on consumables (diesel fuel, battery capacity) must have inventories sufficient for the design-basis duration or be supplemented by passive systems that do not require consumables.

The design-basis duration is facility-class-specific and is set by the governing safety-function timeline for the applicable facility class, not by a single series-wide constant. Two facility-class values are established in the current record and serve as the first instances of this principle.

For large light-water reactor facilities, the design-basis duration shall not be less than two years. This value is derived from the spent fuel pool thermal analysis documented in DBA-MA-UC-1 and from the operational precedent established at Zaporizhzhia Nuclear Power Plant, where occupation has extended beyond that threshold. The two-year floor is the minimum design target; facilities in geopolitically elevated environments may be required by national implementation to exceed it.

For near-term integral light-water small modular reactor facilities, the design-basis duration is 72 hours, a conservative class floor anchored to the post-Fukushima station blackout and FLEX coping benchmark that passive designs are required to meet or exceed, as established in DBA-MA-SMR-FC1. This value reflects the governing safety-function timeline for that facility class and stands on its own facility-class analysis rather than on the large light-water reactor value. Additional facility classes shall establish their design-basis duration through the same methodology: identify the governing safety-function timeline, anchor it to the licensed or benchmark minimum for the class, and state it in the applicable facility-class document and national child document.

National child documents shall specify the minimum design-basis duration applicable within each jurisdiction and shall identify any national regulatory requirements that exceed the international minimum established here.

3.4 Acceptance Criteria

The acceptance criteria for the DBA-MA are expressed in terms of the ASSC. The facility must demonstrate, through analysis, that it can achieve the ASSC from any initial operating condition; complete the transition to ASSC within the timeframes specified in the Trigger Framework under the degraded conditions postulated for Phase 2; maintain the ASSC for the design basis duration using only on-site resources; and maintain off-site dose consequences within the applicable national regulatory dose limits throughout the transition and ASSC phases.

The defining acceptance criterion for the DBA-MA is that the facility's safety functions can be maintained in the Autonomous Safe Shutdown Condition using only systems and resources physically present at the site, independent of any external system that may be compromised by military action. This is the principle of local sovereignty applied as a regulatory acceptance

criterion. The specific dose limit references applicable in each national jurisdiction are identified in the corresponding national child document.

4. Engineering Design Requirements

4.1 Governing Principles

The engineering design requirements for the DBA-MA are governed by seven principles that extend the existing safety design criteria—as established under the IAEA Safety Requirements series and implemented through applicable national regulatory frameworks—to the military action threat profile. These principles apply to all systems, structures, and components credited in the DBA-MA safety analysis.

Local sovereignty. All systems credited for achieving and maintaining the ASSC must be capable of performing their safety function without dependence on any external system—off-site power, supply chains, communications, emergency response, or regulatory direction—that may be compromised by military action. A protection system must maintain its function independent of the compromise of any external system upon which it might otherwise depend. This is the foundational design principle for the DBA-MA.

Extended autonomous operation. Systems credited for ASSC maintenance must be capable of sustained operation for the design basis duration without external consumable resupply. Where this is not achievable through stored consumables alone, passive systems that do not require consumables must be provided as the ultimate backup. The design must demonstrate that time margins increase, rather than decrease, as decay heat declines over the ASSC duration.

Hardening against intentional destructive force. Systems credited for ASSC achievement and maintenance must be protected against the effects of conventional explosive ordnance as characterized in the classified threat assessment. Protection strategies may include physical hardening, burial or bunkering below grade, dispersing redundant components to prevent common-cause failure from a single strike, and separation distances that exceed the damage radius of the postulated ordnance. Specific hardening requirements will be developed in classified supplements to this document.

Degraded-staffing operability. The design must ensure that the ASSC can be achieved and maintained with staffing levels significantly below the normal operating complement. This requires automatic actuation of safety functions by hard-wired or qualified deterministic means to the maximum extent practicable, simplifying ASSC transition and maintenance procedures, and designing features that reduce the frequency and complexity of operator actions required to maintain safe conditions. The design basis staffing level for ASSC operations will be specified in site-specific analyses.

Graceful degradation. Systems credited for ASSC must be designed so that progressive loss of components or functions results in a progressive reduction in capability and margin, rather than an abrupt loss of safety function. The design must avoid cliff-edge effects where the loss of a single component or system causes a step change from safe to unsafe conditions.

Concealment and signature reduction. Systems credited for ASSC achievement and maintenance that cannot be hardened to survive attack by uncrewed aerial systems shall, where practicable, be concealed from aerial observation by below-grade installation, visual barriers or camouflage, and the thermal and electromagnetic signatures of operating equipment shall be assessed and, where practicable, reduced. Concealment complements hardening; it does not replace it. It degrades the adversary's surveillance-to-strike cycle and increases the effort required to achieve a safety-significant effect.

Electromagnetic hardening. Safety-critical instrumentation and control credited for the ASSC shall be evaluated for vulnerability to nuclear electromagnetic pulse and high-power microwave effects and hardened to the threat characterization for the facility's environment, using shielding, surge protection, fiber-optic isolation and separation of redundant channels. Each facility shall maintain analog backup instrumentation and, where practicable, analog or relay-based backup protection for the minimum safety functions required to monitor and maintain the ASSC, independent of digital systems that such effects may disable, and its ASSC procedures shall provide for diagnosing and responding to unexplained failures of instrumentation and control.

These seven principles are those of the parent framework, Sections 5.1 through 5.7. Cyber attack on plant digital systems is prevented and detected under each state's computer security regime; this document requires that the ASSC be achievable and maintainable despite the effects of a successful cyber attack, alone or combined with kinetic action.

4.2 Specific Engineering Requirements

[Note: The following subsections identify the categories of specific engineering requirements to be developed in subsequent revisions. Each subsection will contain detailed design requirements derived from the governing principles and the postulated accident conditions.]

4.2.1 Electrical Power Systems

Requirements for on-site electrical power systems capable of supporting ASSC operations for the design basis duration, including diesel generator capacity and fuel storage, battery systems, passive power sources (if applicable), and protection of electrical distribution systems against ordnance damage. Requirements must address the sustained, deliberate disruption of off-site transmission infrastructure and the potential for explosive ordnance to damage on-site electrical systems.

4.2.2 Decay Heat Removal Systems

Requirements for decay heat removal systems capable of transitioning from active cooling to increasingly passive modes as decay heat declines, to achieve fully passive decay heat removal within the ASSC duration. Requirements must address both reactor core and spent fuel pool cooling and account for the possibility that primary or secondary cooling system piping may sustain damage from ordnance.

4.2.3 Containment Systems

Requirements for containment integrity under the postulated ordnance loading, including structural hardening, protection of containment penetrations, and provisions for containment isolation under degraded conditions. Requirements must account for the possibility that containment integrity must be maintained without active leak detection, pressure management, or maintenance for the duration of the ASSC.

4.2.4 Instrumentation and Control

Requirements for essential instrumentation capable of sustained autonomous operation, including power supply, environmental qualification for post-ordnance conditions, and design for extended operation without calibration or maintenance. Requirements must specify the minimum instrumentation required to confirm that ASSC conditions are maintained, and must implement the electromagnetic hardening and analog backup provisions of Section 4.1.

4.2.5 Physical Protection of Safety Systems

Requirements for the physical protection of all systems, structures, and components credited in the DBA-MA safety analysis against the effects of explosive ordnance and hostile action. This includes hardening specifications, separation and dispersal requirements, requirements for below-grade or bunkered installations, and passive kinetic standoff barrier systems for components that cannot be practicably hardened, buried, or concealed but remain credited in the ASSC safety analysis. Passive kinetic standoff barriers enforce a detonation distance between UAS-delivered warheads and protected assets, converting precision-strike effects into attenuated area-effect loading at a non-optimal range; their design basis, threat-class scaling, and installation requirements are addressed in DBA-UxS (Section 8.2) and DBA-MA-UxS-NA. Specific threat characterizations and protection specifications will be developed in classified supplements.

4.2.6 Consumables and Stored Resources

Requirements for on-site storage of all consumables necessary for ASSC operations for the design basis duration, including diesel fuel, lubricating oil, water treatment chemicals, passive kinetic defense barrier replacement materials (mesh, cable, structural hardware, and fasteners sufficient to restore all design-basis barrier installations following a strike event), and other materials. Requirements must address protection of consumable storage against ordnance damage and environmental degradation over the ASSC duration.

5. Operational Requirements

5.1 Procedures

Each facility shall develop, maintain, and periodically exercise procedures for each phase of the DBA-MA: Phase 1 preparation activities, including consumable inventory management, system readiness verification, staff training, and coordination with external authorities; Phase 2 transition to the ASSC, including controlled shutdown from any operating state, system alignment for autonomous operation, and staffing transition; Phase 3 ASSC maintenance, including routine monitoring, consumable management, and contingency actions for equipment degradation; and Phase 4 recovery assessment and return-to-service evaluation. These procedures shall be integrated with the facility's existing emergency operating procedures and severe accident management guidelines.

5.2 Training

Operating staff shall receive training on the DBA-MA phases, the ASSC concept, transition procedures, and ASSC maintenance operations. Training shall include exercises conducted at intervals determined by the applicable national regulatory authority, with scenarios that address the specific challenges of the military action: degraded staffing, loss of communications, extended autonomous operations, and decision-making under the Phase 1 operational dilemma.

5.3 Trigger Framework Implementation

Each facility shall implement the Trigger Framework (DBA-MA L3) as specified in the companion document and as adapted in the applicable national child document. This includes establishing communication channels, information access agreements, and decision-authority frameworks necessary to receive and act on the trigger indicators specified in the Trigger Framework. The Trigger Framework implementation shall be verified through periodic exercises conducted in coordination with the relevant intelligence, military, and emergency management authorities.

6. References and Related Documents

6.1 DBA-MA Series Documents

DBA-MA International Framework (L1) — Expanding the Design Basis for Nuclear Power Plants to Address Military Action Against Nuclear Facilities: An International Framework. Eclectic Technologies.

DBA-MA L3 — Trigger Framework for Military Action Design Basis Accident: Observable Indicators and Decision Criteria for Transition to Autonomous Safe Shutdown Condition. Eclectic Technologies.

DBA-MA L4 — Monitoring Sources and Data-Sharing Guidance for DBA-MA Trigger Observables. Eclectic Technologies.

DBA-UxS — Uncrewed Systems: Threat Considerations for Critical Infrastructure. Eclectic Technologies.

DBA-MA-UxS-NA — Uncrewed Systems: Nuclear Sector Implementation Annex. Eclectic Technologies.

DBA-MA-SMR-FC1 — SMR Facility-Class Design Basis. Eclectic Technologies.

DBA-MA-UC-1 — Use Case: Spent Fuel Pool Thermal Progression. Eclectic Technologies.

6.2 IAEA References

IAEA Safety Fundamentals No. SF-1 — Fundamental Safety Principles (2006).

IAEA Safety Standards Series No. SSR-2/1 — Safety of Nuclear Power Plants: Design (Rev. 1, 2016).

IAEA Safety Standards Series No. GSR Part 4 (Rev. 1) — Safety Assessment for Facilities and Activities (2016).

IAEA Nuclear Security Series No. 10-G (Rev. 1) — National Nuclear Security Threat Assessment, Design Basis Threats and Representative Threat Statements (2021).

IAEA Nuclear Security Series No. 42-G — Computer Security for Nuclear Security (2021).

IAEA, Nuclear Safety, Security and Safeguards in Ukraine — Director General summary reports and updates (2022–present), including the 2nd Summary Report, 28 April – 5 September 2022.

6.3 National Implementation References

National regulatory and legislative references applicable to each implementation of this document are identified in the corresponding national child document. The United States national references are set out in DBA-MA L2, United States National Implementation. References for Finland, and for other national implementations as they are developed, are set out in the respective national child documents.

Revision History

Revision notes for versions earlier than 3.0 were not carried in the drafts and are not reconstructed here.

Revision Note (v3.0, September 2026): Issued as the first final version, aligned with the DBA-MA International Framework, Version 2.0. Removes the draft status and adds the citation block, stamped with the reserved version DOI 10.5281/zenodo.22945300. States the design-basis method and design-extension-condition placement (Section 1.3), points facility classes to DBA-MA-SMR-FC1 (Section 1.2), and maps the accident description to the parent's six threat categories (Section 2.1). Adds the principles of concealment and signature reduction and of electromagnetic hardening, bringing the governing principles to seven as in the parent, with a statement on cyber attack; qualifies automation as hard-wired or qualified deterministic; carries electromagnetic provisions into Section 4.2.4. Corrects the subcriticality means in Section 3.2, restates the Section 3.3 minimum-duration sentence, and aligns the small modular reactor duration with DBA-MA-SMR-FC1 as a 72-hour class floor. Corrects the DBA-UxS section reference to Section 8.2, cites series instruments by designation without version numbers, corrects the GSR Part 4 (Rev. 1) title and year, and names the Nuclear Security Series documents relied on.